


# **L10b**: The NPV of Composite Contracts and A New Role for Delta

CHEME 5660 Cornell University

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# Objectives for Today

First, two examples on the Greeks, the POE, and the POP. Then: what is a composite contract worth, and when does it make money?

## Today's concepts

- The NPV and the probability of profit of a composite contract: sums over its legs.
- Common composite contracts at expiration: outlook, breakevens, maximum profit and loss, and the cost of closing early.
- Delta-neutral positions: the net Delta of options and shares, and the shares that make it zero.

# What You Need to Know

## Core results

- Profit at expiration: the legs' payoffs, weighted by $\theta_i n_i$, minus the composite premium grown to expiration. POP: the probability it is positive.
- Read a profit diagram: the breakevens, the maximum profit (for the eight credit contracts, the credit grown to expiration), and the maximum loss (from the put or call side).
- Composite Delta: $\Delta_{\mathcal{C}} = \sum_{i \in \mathcal{C}} \theta_i n_i \Delta_i$ (per share). Options' Delta: $\Delta_{\text{opts}} = m\,\Delta_{\mathcal{C}}$. Holding $N_S = -\Delta_{\text{opts}}$ shares makes the net Delta, $N_S + \Delta_{\text{opts}}$, zero.

## For understanding (you won't reproduce these)

- The NPV derivation, and the formula tables for the eight composite contracts (reference).

# Lectures and Examples

**Lecture:**
CHEME-5660-L10b-Lecture-POP-POE-Delta-CompositeContracts-Fall-2026.ipynb

**Example 1:**
CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

**Example 2:**
CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

**Example 3:**
CHEME-5660-L10b-Example-Delta-VerticalSpread-Fall-2026.ipynb

**Review on your own:**
CHEME-5660-L10b-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

# Review: The Greeks

For small changes in the share price, the volatility, the calendar time (years), and the rate, the premium $\mathcal{P}_\star$ of a call ($\star = c$) or put ($\star = p$) changes by:

$$\begin{aligned}\Delta\mathcal{P}_\star &\approx \Delta_\star\,\Delta S + \tfrac{1}{2}\,\Gamma_\star\,(\Delta S)^2 + \nu_\star\,\Delta\sigma + \Theta_\star\,\Delta t + \rho_\star\,\Delta g_y \\ &= \underbrace{(\Delta_\star + \tfrac{1}{2}\,\Gamma_\star\,\Delta S)\Delta S}_{\text{share price}} + \underbrace{\nu_\star\,\Delta\sigma}_{\text{volatility}} + \underbrace{\Theta_\star\,\Delta t}_{\text{time}} + \underbrace{\rho_\star\,\Delta g_y}_{\text{rate}}\end{aligned}$$

- Delta and Gamma: the slope and curvature of the premium in $S$.
- Vega, Theta, and Rho: sensitivities to the volatility $\sigma$, the calendar time $t$, and the risk-free rate $g_y$, quoted as $\Theta_\star/365$ per day and $\nu_\star/100$, $\rho_\star/100$ per point.
- Hedge ratio: one contract covers 100 shares, so it is equivalent to $100 \cdot \Delta_\star$ shares.

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**Example 1:** CHEME-5660-L10a-Example-Greeks-PutCall-Fall-2026.ipynb

# Review: POE and POP

The POE is the probability of finishing in the money by more than 0.01 USD/share; the POP is the probability of a positive profit at expiration, beyond the breakeven $S^{\ddagger} = K \pm \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_\star$ ($+$ call, $-$ put):

$$\text{POE} = \begin{cases} \mathbb{P}\left(S(T) > K + 0.01\right) & \text{call} \\ \mathbb{P}\left(S(T) < K - 0.01\right) & \text{put} \end{cases}$$

$$\text{POP} = \begin{cases} \mathbb{P}\left(S(T) > S^{\ddagger}\right) & \text{bought a call or sold a put} \\ \mathbb{P}\left(S(T) < S^{\ddagger}\right) & \text{sold a call or bought a put} \end{cases}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$; we simulate $S(T)$ under our belief $\mathbb{P}$. However, the POE $\sim |\Delta_\star|$, the Delta quoted by trading platforms under $\mathbb{Q}$.

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**Example 2:** CHEME-5660-L10a-Example-POE-POP-Call-Fall-2026.ipynb

# Composite Contracts

A composite contract $\mathcal{C}$ combines long and short puts and calls on one underlying asset with one expiration date $T$. Its payoff per share at expiration is:

$$\hat{V}_{\mathcal{C}}(S(T)) = \underbrace{\sum_{i \in \mathcal{C}} \theta_i \, n_i \, \overbrace{V_i(K_i, S(T))}^{\text{payoff of leg } i}}_{\text{sum of all legs}}$$

Here $S(T)$ is the share price at expiration, $K_i$ the strike price of leg $i$, $V_i$ its payoff, $(S(T) - K_i)^+$ for a call or $(K_i - S(T))^+$ for a put, $n_i$ its number of contracts, and $\theta_i = -1$ if short or $1$ if long.

# The Net Present Value of a Composite Contract

A composite contract is a sum of legs, so its NPV is the sum of the legs' NPVs, weighted by $\theta_i n_i$:

$$\mathrm{NPV}_0(\mathcal{C}) = \sum_{i\in\mathcal{C}} \theta_i\, n_i \underbrace{\left(-\mathcal{P}_i + \mathcal{D}_{T,0}^{-1}(g_y)\cdot V_i(K_i, S(T))\right)}_{\text{NPV of buying leg } i} \qquad \text{weighted leg NPVs}$$

$$= -\underbrace{\sum_{i\in\mathcal{C}} \theta_i\, n_i\, \mathcal{P}_i}_{=\,\mathcal{P}_{\mathcal{C}}} + \mathcal{D}_{T,0}^{-1}(g_y)\cdot \underbrace{\sum_{i\in\mathcal{C}} \theta_i\, n_i\, V_i(K_i, S(T))}_{=\,\hat{V}_{\mathcal{C}}(S(T))} \qquad \text{split the sum}$$

$$= -\mathcal{P}_{\mathcal{C}} + \mathcal{D}_{T,0}^{-1}(g_y)\cdot \hat{V}_{\mathcal{C}}(S(T))$$

where $\mathcal{P}_i$ is the premium of leg $i$ (USD/share) and $\mathcal{P}_{\mathcal{C}}$ is the composite premium.

# Profit at Expiration

Factoring out $\mathcal{D}_{T,0}^{-1}(g_y)$ writes the NPV as a profit at expiration discounted to today:

$$\begin{aligned}\mathrm{NPV}_0(\mathcal{C}) &= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \underbrace{\left(\hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}\right)}_{= \hat{P}_{\mathcal{C}}(S(T))} \qquad \text{factor out } \mathcal{D}_{T,0}^{-1}(g_y) \\ &= \mathcal{D}_{T,0}^{-1}(g_y) \cdot \hat{P}_{\mathcal{C}}(S(T))\end{aligned}$$

where $\mathcal{D}_{T,0}(g_y) = e^{g_y T}$ is the accumulation factor, and:

$$\boxed{\hat{P}_{\mathcal{C}}(S(T)) = \hat{V}_{\mathcal{C}}(S(T)) - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_{\mathcal{C}}}$$

is the profit at expiration (USD/share). $\mathcal{P}_{\mathcal{C}} > 0$ opens for a debit; $\mathcal{P}_{\mathcal{C}} < 0$ opens for a credit of $-\mathcal{P}_{\mathcal{C}}$.

# Probability of Profit and Probability of Exercise

The NPV and the profit at expiration have the same sign, so with $\mathbb{P}$ our belief, the probability of profit is:

$$\boxed{\text{POP}_{\mathcal{C}} = \mathbb{P}\Big(\text{NPV}_0(\mathcal{C}) > 0\Big) = \mathbb{P}\Big(\hat{P}_{\mathcal{C}}(S(T)) > 0\Big)}$$

Each leg has its own probability of exercise:

$$\text{POE}_i = \begin{cases} \mathbb{P}\left(S(T) > K_i + 0.01\right) & \text{if leg } i \text{ is a call} \\ \mathbb{P}\left(S(T) < K_i - 0.01\right) & \text{if leg } i \text{ is a put} \end{cases}$$

# Common Composite Contracts

Each contract below opens for a credit and has a put side and a call side:

- Sides: empty, a naked short contract, or a credit spread (a short contract protected by a long one farther out).
- Outlook: the put side loses if the price falls, the call side if it rises. Only the put side: neutral to bullish; only the call side: neutral to bearish; both: neutral.

# Common Composite Contracts: Sides and Outlook

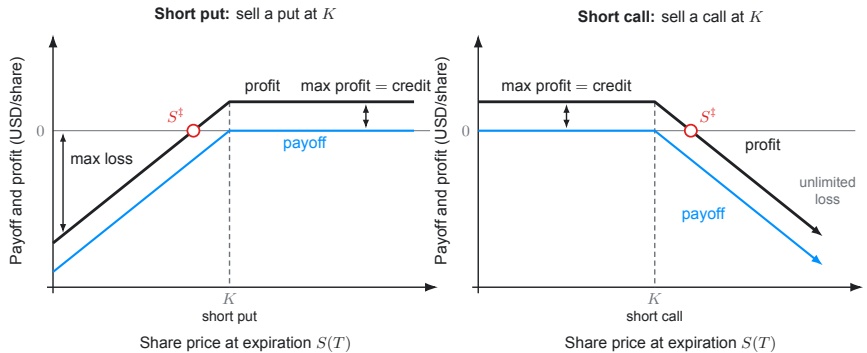
The eight structures, by side:

| Composite contract | Put side | Call side | Outlook |
|---|---|---|---|
| Short put | naked short put | none | neutral to bullish |
| Short call | none | naked short call | neutral to bearish |
| Put credit spread | put credit spread | none | neutral to bullish |
| Call credit spread | none | call credit spread | neutral to bearish |
| Short straddle | naked short put | naked short call at the same strike | neutral |
| Short strangle | naked short put | naked short call at a higher strike | neutral |
| Protected strangle | naked short put | call credit spread | neutral, leaning bullish |
| Iron condor | put credit spread | call credit spread | neutral |

Strikes are numbered low to high, $\mathcal{P}_i$ is the premium of the leg at $K_i$. Figures: payoff (blue), profit (black).

# Short Puts and Short Calls

Selling one put, a short put, or one call, a short call, is the building block of every structure below:




Each keeps the whole premium if it expires out of the money.

# Short Puts and Short Calls: Profit at Expiration

With premiums $\mathcal{P}_p$ (put) and $\mathcal{P}_c$ (call) at strike $K$, the profits at expiration, and the breakevens $S^{\ddagger}$, are:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p - (K - S(T))^+ \qquad \text{short put}$$

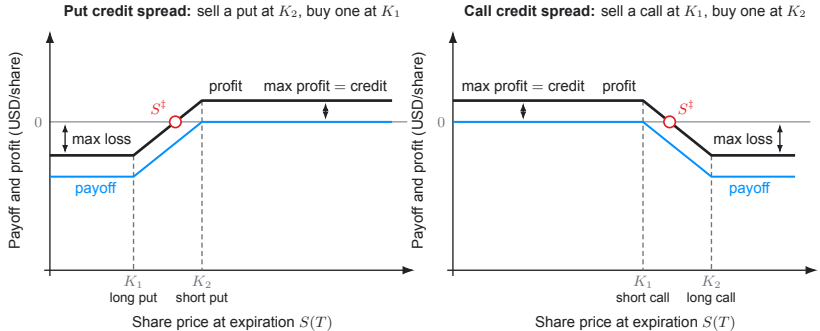
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c - (S(T) - K)^+ \qquad \text{short call}$$

| | Short put | Short call |
|---|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ for $S(T) \geq K$ | $\mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$ for $S(T) \leq K$ |
| Breakeven, put side | $S^{\ddagger} = K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ | none |
| Maximum loss, put side | $K - \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_p$ at $S(T) = 0$ | none |
| Breakeven, call side | none | $S^{\ddagger} = K + \mathcal{D}_{T,0}(g_y) \cdot \mathcal{P}_c$ |
| Maximum loss, call side | none | unlimited |

The short put's loss is bounded ($S(T) \geq 0$); the short call's is not.

# Credit Spreads

A credit spread sells the contract closer to the share price and buys one farther away, same type and expiration (a bull put or bear call spread):




The short leg collects the credit; the long leg caps the loss.

# Put Credit Spread

A long put at $K_1$ and a short put at $K_2$: the credit grown to expiration, minus and plus the leg payoffs:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1) - (K_2 - S(T))^+ + (K_1 - S(T))^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \geq K_2$ |
| Breakeven, put side | $S^{\ddagger} = K_2 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ |
| Maximum loss, put side | $(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 - \mathcal{P}_1)$ for $S(T) \leq K_1$ |

# Call Credit Spread

A short call at $K_1$ and a long call at $K_2$ give the profit at expiration:

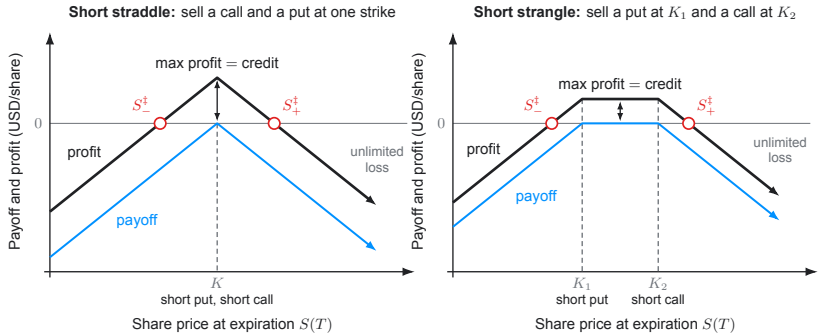
$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2) - (S(T) - K_1)^+ + (S(T) - K_2)^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \leq K_1$ |
| Breakeven, call side | $S^{\ddagger} = K_1 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ |
| Maximum loss, call side | $(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 - \mathcal{P}_2)$ for $S(T) \geq K_2$ |

Beyond the long strike, the long leg gains what the short leg loses, so the loss stops at the strike width minus the credit.

# Short Straddles and Strangles

A straddle: a put and a call at one strike $K$. A strangle: a put at $K_1$ below a call at $K_2$. Same expiration:




Sold, each profits only between its breakevens: POP $= \mathbb{P}(S_-^\ddagger < S(T) < S_+^\ddagger)$.

# Short Straddle

Selling a put and a call at the strike $K$ gives the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c) - (K - S(T))^+ - (S(T) - K)^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = K$ |
| Breakeven, put side | $S_-^\ddagger = K - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ |
| Maximum loss, put side | $K - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ at $S(T) = 0$ |
| Breakeven, call side | $S_+^\ddagger = K + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_p + \mathcal{P}_c)$ |
| Maximum loss, call side | unlimited |

The straddle collects more credit, but the strangle keeps its credit over a wider range of share prices.

# Short Strangle

Selling a put at $K_1$ and a call at $K_2$ gives the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2) - (K_1 - S(T))^+ - (S(T) - K_2)^+$$

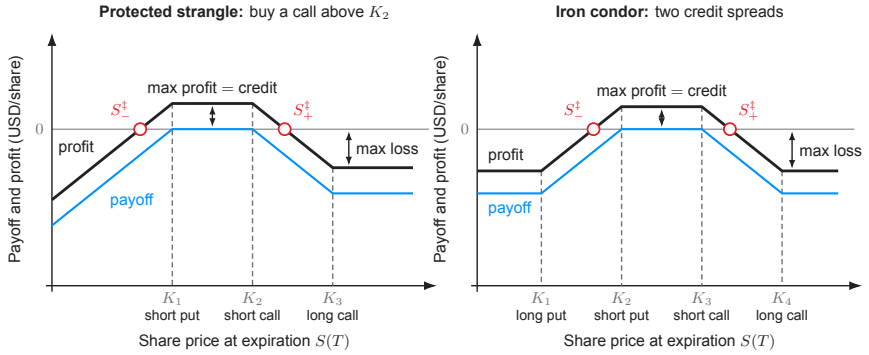
| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ for $K_1 \leq S(T) \leq K_2$ |
| Breakeven, put side | $S_-^\ddagger = K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ |
| Maximum loss, put side | $K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ at $S(T) = 0$ |
| Breakeven, call side | $S_+^\ddagger = K_2 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2)$ |
| Maximum loss, call side | unlimited |

Buying both legs ($\theta_i = 1$) flips either diagram: the loss is capped at the premium paid, and a large move either way pays.

**Review on your own:** CHEME-5660-L10b-Example-Straddle-AMD-Earnings-Fall-2026.ipynb

# Protected Strangles and Iron Condors

A protected strangle (jade lizard) adds a long call at $K_3$ to a short strangle; an iron condor also adds a long put:




Each long leg caps the loss on its side but lowers the credit.

# Protected Strangle

A short put at $K_1$, a short call at $K_2$, and a long call at $K_3$ give the profit at expiration:

$$\hat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) - (K_1 - S(T))^+ - (S(T) - K_2)^+ + (S(T) - K_3)^+$$

protected strangle

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ for $K_1 \leq S(T) \leq K_2$ |
| Breakeven, put side | $S_-^\ddagger = K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ |
| Maximum loss, put side | $K_1 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ at $S(T) = 0$ |
| Breakeven, call side | $S_+^\ddagger = K_2 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$[a] |
| Maximum loss, call side | $(K_3 - K_2) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3)$ for $S(T) \geq K_3$[b] |

[a] If $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) \leq K_3 - K_2$; otherwise none. [b] If $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_1 + \mathcal{P}_2 - \mathcal{P}_3) < K_3 - K_2$; otherwise none.

If the credit grown to expiration is at least $K_3 - K_2$, the call side cannot lose.

# Iron Condor

A long put at $K_1$, a short put at $K_2$, a short call at $K_3$, and a long call at $K_4$ give the profit at expiration:

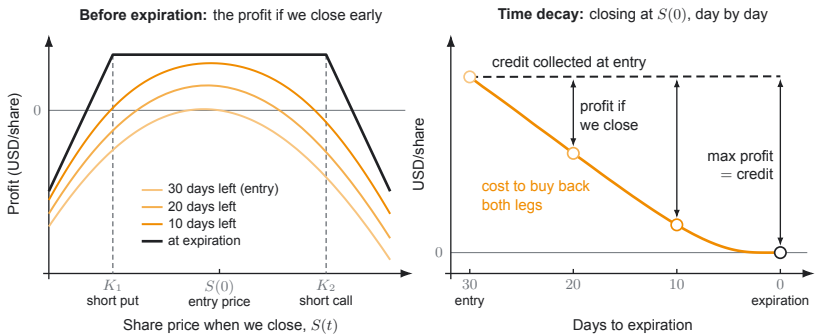
$$\widehat{P}_{\mathcal{C}}(S(T)) = \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4) + (K_1 - S(T))^+ - (K_2 - S(T))^+ - (S(T) - K_3)^+ + (S(T) - K_4)^+$$

| | |
|---|---|
| Maximum profit | $\mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $K_2 \le S(T) \le K_3$ |
| Breakeven, put side | $S_-^\ddagger = K_2 - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ |
| Maximum loss, put side | $(K_2 - K_1) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \le K_1$ |
| Breakeven, call side | $S_+^\ddagger = K_3 + \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ |
| Maximum loss, call side | $(K_4 - K_3) - \mathcal{D}_{T,0}(g_y) \cdot (\mathcal{P}_2 + \mathcal{P}_3 - \mathcal{P}_1 - \mathcal{P}_4)$ for $S(T) \ge K_4$ |

At any share price at expiration, at most one side can lose, so the maximum loss is the wider strike width minus the credit.

# Why Wait to Expiration?

Our diagrams are drawn at expiration. Closing a short leg early means buying it back: intrinsic plus extrinsic value $E_V(t)$. A short strangle (put $K_1 = 90$, call $K_2 = 110$, $S(0) = 100$ USD/share, 30 days, CRR tree, $\sigma = 0.40$, $g_y = 4\%$):



# Why Wait to Expiration? What Do We See?

- Before expiration (left): the buyback price still includes extrinsic value, so earlier curves lie below the profit at expiration.
- At the entry share price (right): both legs are out of the money, so closing at once costs exactly the credit and earns nothing.
- Time decay: as the extrinsic value decays, the profit if we close at $S(0)$ rises to the whole credit.

The diagrams show the most each contract can earn; an early exit earns less, by roughly the extrinsic value left in the legs.

# Delta as a Hedging Tool for Composite Contracts

Delta is also the number of shares that hedges one contract. A composite position's Delta is the sum of its legs' Deltas, so the same idea hedges the whole position.

**What is hedging?** Offsetting potential losses with an opposite position in a related counter asset, often the underlying shares.

A position with zero Delta is insensitive to small changes in the share price.

# What Is the Delta of a Composite Contract?

We hold $\theta_i n_i$ American contracts of leg $i$, each for $m$ shares, and $N_S$ shares. With $\mathcal{P}_i(S,t)$ the premium of leg $i$ at share price $S$ and time $t$, the portfolio is worth $\Pi(S,t) = N_S S + m \sum_{i\in\mathcal{C}} \theta_i n_i \mathcal{P}_i(S,t)$, and the net Delta is its sensitivity to the share price:

$$\Delta_{\text{net}}(S,t) = \frac{\partial \Pi}{\partial S} = \underbrace{\frac{\partial (N_S S)}{\partial S}}_{=N_S} + m \sum_{i\in\mathcal{C}} \theta_i n_i \underbrace{\frac{\partial \mathcal{P}_i}{\partial S}}_{=\Delta_i(S,t)}$$

$$= N_S + \underbrace{m \sum_{i\in\mathcal{C}} \theta_i n_i \Delta_i(S,t)}_{=\Delta_{\text{opts}}(S,t)}$$

The options contribute $\Delta_{\text{opts}} = m\,\Delta_{\mathcal{C}}$, where $\Delta_{\mathcal{C}}(S,t) = \sum_{i\in\mathcal{C}} \theta_i n_i \Delta_i(S,t)$ is the composite Delta, per share.

# Delta-Neutral Position

A position is delta-neutral if $\Delta_{\text{net}}(S,t) = 0$. We get there by choosing $N_S$ to offset the Delta of the options:

$$\Delta_{\text{net}}(S,t) = N_S + \Delta_{\text{opts}}(S,t) = 0 \quad \Rightarrow \quad N_S = -\Delta_{\text{opts}}(S,t)$$

**Quick gut-check example:** short $10$ calls at $\Delta = 0.6$ with $m = 100$ shares/contract, so $\Delta_{\text{opts}} = 10 \times (-1) \times 0.6 \times 100 = -600$. Holding $N_S = +600$ shares gives $\Delta_{\text{net}} = 600 - 600 = 0$.

If leg $i$ is exercised or assigned at time $\tau \leq T$, set $\Delta_i = 0$ for $t \geq \tau$ and update the shares:

- Call leg $i$: $N_S \leftarrow N_S + \theta_i \cdot m \cdot n_i$.
- Put leg $i$: $N_S \leftarrow N_S - \theta_i \cdot m \cdot n_i$.

# Delta of the Common Composite Contracts

Short legs at Deltas $\mp 0.25$, long legs at $\mp 0.10$ beyond them, and a straddle at $\mp 0.5$. At entry, one copy of each leg ($m = 100$):

| Composite contract | Composite Delta, $\Delta_c$ | Shares to hedge, $N_S$ |
|---|---|---|
| Short put | $-(-0.25) = 0.25$ | $-25$ |
| Short call | $-0.25$ | $+25$ |
| Put credit spread | $-0.10 - (-0.25) = 0.15$ | $-15$ |
| Call credit spread | $-0.25 + 0.10 = -0.15$ | $+15$ |
| Short straddle | $-(-0.5) - 0.5 \approx 0$ | $\approx 0$ |
| Short strangle | $-(-0.25) - 0.25 = 0$ | $0$ |
| Protected strangle | $-(-0.25) - 0.25 + 0.10 = 0.10$ | $-10$ |
| Iron condor | $-0.10 - (-0.25) - 0.25 + 0.10 = 0$ | $0$ |

At these strikes, the sign of $\Delta_c$ matches the outlook.

# Reading the Delta Table

$\Delta_{\mathcal{C}}$ is the position's gain per share when the share price rises 1 USD/share:

- The rule of thumb: a Delta of $\pm 0.25$ is roughly a one-in-four chance of finishing in the money under $\mathbb{Q}$.
- The protected strangle: its long call leaves $+0.10$, so it leans bullish.
- Zero is not zero risk: Gamma pulls the Delta away from zero as the share price moves.

Staying delta-neutral means updating $N_S$ as the Deltas change.

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**Example 3:** CHEME-5660-L10b-Example-Delta-VerticalSpread-Fall-2026.ipynb

# Summary

## Key takeaways

- The NPV of a composite contract is the sum of its legs' NPVs: the POP is the probability that it is positive.
- Common composite contracts have a put side and a call side: each side is empty, naked, or a credit spread, which sets the outlook and the maximum loss.
- Delta builds hedged positions: holding $N_S = -\Delta_{\text{opts}}$ shares makes the net Delta zero.